

Exp No: 5

ARDUINO BASED MANUAL ELECTRONIC COUNTER USING

Program;

```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;

int delay_time=200;

void setup() {

  //configure pin2 as an input and enable the internal pull-up resistor
  pinMode(12, INPUT_PULLUP);

  pinMode(1,OUTPUT);
  pinMode(2,OUTPUT);
  pinMode(3,OUTPUT);
  pinMode(4,OUTPUT);
  pinMode(5,OUTPUT);
  pinMode(6,OUTPUT);
  pinMode(7,OUTPUT);
}

void loop() {

  //read the pushbutton value into a variable
  int sensorVal = digitalRead(12);

  if (sensorVal == LOW) {

    x0=true;

  }

  while(x0){

    zero();

    sensorVal = digitalRead(12);

    if (sensorVal == LOW) {

      x1=true;

      x0=false;

    }

  }

  while(x1){

    one();

    sensorVal = digitalRead(12);

    if (sensorVal == LOW) {

      x2=true;

      x1=false;
```

```
}  
  
}  
  
while(x2){  
  
two();  
  
sensorVal = digitalRead(12);  
  
if (sensorVal == LOW) {  
  
x3=true;  
  
x2=false;  
  
}  
  
}  
  
while(x3){  
  
three();  
  
sensorVal = digitalRead(12);  
  
if (sensorVal == LOW) {  
  
x4=true;  
  
x3=false;  
  
}  
  
}  
  
while(x4){  
  
four();  
  
sensorVal = digitalRead(12);  
  
if (sensorVal == LOW) {  
  
x5=true;  
  
x4=false;  
  
}  
  
}  
  
while(x5){  
  
five();  
  
sensorVal = digitalRead(12);  
  
if (sensorVal == LOW) {x6=true;  
  
x5=false;  
  
}  
  
}  
  
while(x6){  
  
six();  
  
sensorVal = digitalRead(12);  
  
if (sensorVal == LOW) {
```

```
x7=true;
x6=false;
}
}
while(x7){
seven();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x8=true;
x7=false;
}
}
while(x8){
eight();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x9=true;
x8=false;
}
}
while(x9){
nine();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x0=true;
x9=false;
}
}
}
void zero()
{
for(int i=1;i<7;i++)
{
digitalWrite(i,HIGH);
digitalWrite(7,LOW);
}
delay(delay_time);
```

```
}  
  
void one()  
{  
    digitalWrite(1,LOW);  
    digitalWrite(2,HIGH);  
    digitalWrite(3,HIGH);  
    digitalWrite(4,LOW);  
    digitalWrite(5,LOW);  
    digitalWrite(6,LOW);  
    digitalWrite(7,LOW);  
    delay(delay_time);  
}  
  
void two()  
{  
    digitalWrite(1,HIGH);  
    digitalWrite(2,HIGH);  
    digitalWrite(3,LOW);  
    digitalWrite(4,HIGH);  
    digitalWrite(5,HIGH);  
    digitalWrite(6,LOW);  
    digitalWrite(7,HIGH);  
    delay(delay_time);  
}  
  
void three()  
{  
    digitalWrite(1,HIGH);  
    digitalWrite(2,HIGH);  
    digitalWrite(3,HIGH);  
    digitalWrite(4,HIGH);  
    digitalWrite(5,LOW);  
    digitalWrite(6,LOW);  
    digitalWrite(7,HIGH);  
    delay(delay_time);  
}  
  
void four()  
{  
    digitalWrite(1,LOW);  
    digitalWrite(2,HIGH);
```

```
digitalWrite(3,HIGH);digitalWrite(4,LOW);  
digitalWrite(5,LOW);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}  
  
void five()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,LOW);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,LOW);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}  
  
void six()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,LOW);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}  
  
void seven()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,LOW);  
digitalWrite(5,LOW);  
digitalWrite(6,LOW);  
digitalWrite(7,LOW);  
delay(delay_time);
```

```

}

void eight()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}

void nine()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}

```

Output;

